



# Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** across multiple categories.

# Dataset at a Glance

**3,900**

**Transactions**

**18**

**Features**

**\$59.76**

**Avg. Purchase**

**3.75**

**Avg. Rating**

Demographics · Purchase details · Shipping · Discounts · Subscription status

# Data Preparation in Python

01

## Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

## Clean & Impute

37 missing `review_rating` values filled with category median

03

## Feature Engineering

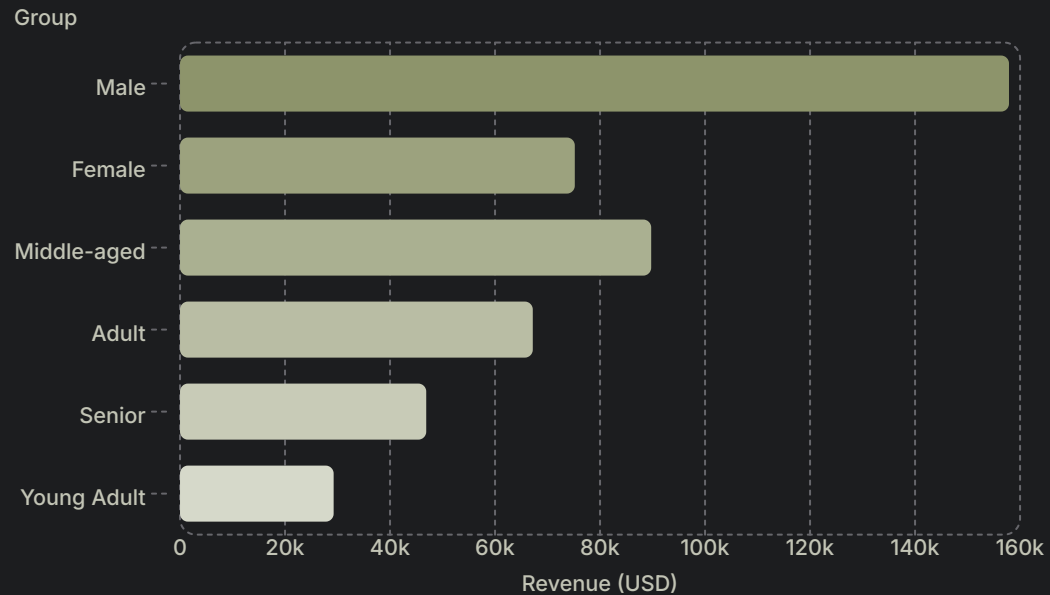
Created `age_group` and `purchase_frequency_days` columns

04

## Load to SQL Server

Cleaned DataFrame exported to SQL Server for structured analysis

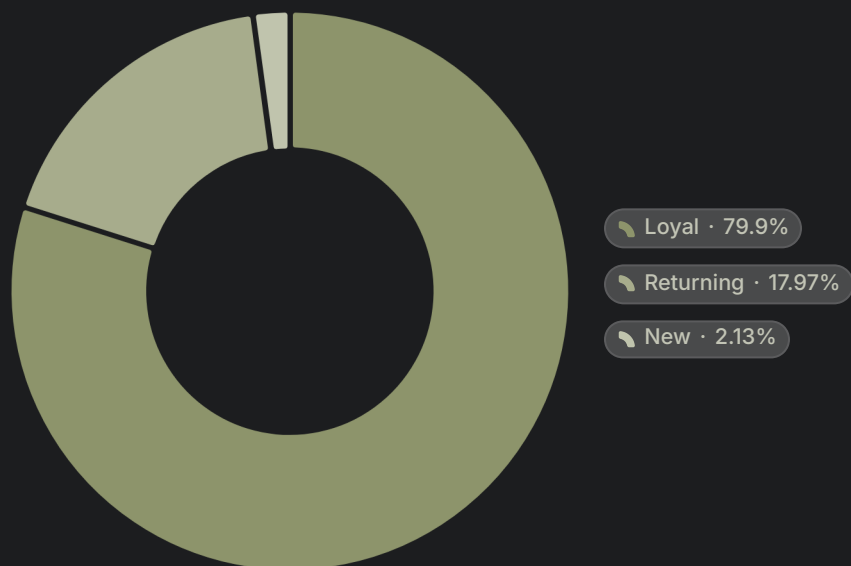
# Revenue Insights



## Key Findings

- Male customers generate **2× more revenue** than female
- Middle-aged segment leads all age groups at **\$89,741**
- Young Adults contribute the least at **\$29,258**

# Customer Segmentation



## Segments by Purchase History

- **Loyal (3,116)** — majority of the base
- **Returning (701)** — re-engaged buyers
- **New (83)** — acquisition opportunity

✓ 958 repeat buyers (>5 purchases) hold active subscriptions

# Subscriptions & Discounts

## Subscribers

27% subscribed · 73% not subscribed

Avg. spend equal at **\$59** for both groups

## Discount-Heavy Products

Hat **50%** · Coat **49%** · Sneakers **49%** ·  
Sweater **48%** · Pants **47%**

## High-Spend Discount Users

Many discount users still spend **above average** — discounts don't erode value

# Top Products & Categories

## Top-Rated Products

| Product | Avg Rating |
|---------|------------|
| Gloves  | 3.86       |
| Sandals | 3.84       |
| Boots   | 3.82       |
| Hat     | 3.80       |
| Skirt   | 3.78       |

## Best Sellers by Category

| Category    | #1 Product           |
|-------------|----------------------|
| Clothing    | Blouse / Pants (171) |
| Accessories | Jewelry (171)        |
| Outerwear   | Jacket (163)         |
| Footwear    | Sandals (160)        |

# CUSTOMER BEHAVIOR DASH



## Power BI Dashboard



### Interactive Filters

Subscription, Gender, Category, Shipping Type



### Revenue & Sales

By category and age group



### Subscription Split

27% Yes · 73% No



# Business Recommendations

## Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscribers

## Loyalty Programs

Reward repeat buyers to grow the Loyal segment

## Review Discount Policy

Balance sales boosts with margin control on high-discount items

## Targeted Marketing

Focus on middle-aged, high-revenue segments and express-shipping users

## Product Positioning

Highlight top-rated and best-selling products in campaigns